AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2012

Biology Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

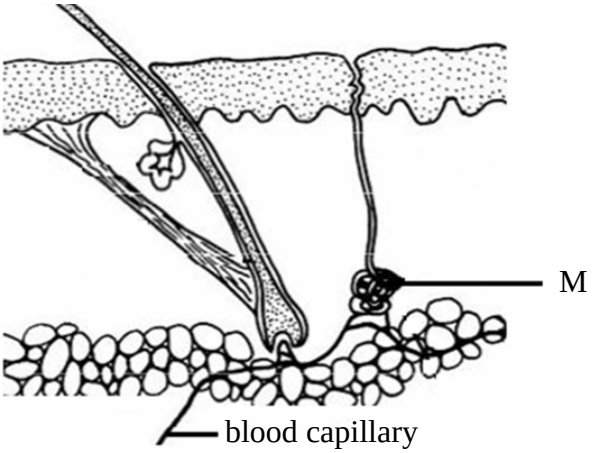
4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. How is photosynthesis different from respiration? (2 Marks)

	Photosynthesis	Respiration
1		
2		

b. The diagram shows a section of the human skin. (2 Marks)



Briefly describe the role of the labelled structure M in keeping the body temperature constant.

_____

_____

_____

_____

Q.2. (Total 4 Marks)

a. What features of nervous control make it better over hormonal control? (2 Marks)

1. _____

_____

2. _____

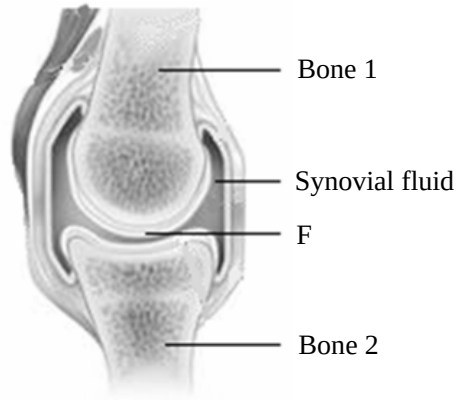
_____

b. Recall your knowledge about a brain disorder called epilepsy and complete the given table. (2 Marks)

Symptoms	1.
	2.
Treatment	

Q.3. (Total 5 Marks)

a. The diagram shows a joint of the skeletal system. (3 Marks)



Identify structure F and describe the disorder which might occur if this structure is degenerated.

b. Why is water necessary for the germination of seeds? (2 Marks)

Q.4. (Total 4 Marks)

a. Human blood is classified into A, B, AB and O blood groups. Out of these, which blood group is an example of co-dominance? Justify your answer with a suitable reason. (2 Marks)

_____

_____

_____

_____

b. Variations are differences in traits or characteristics between individuals of the same species. How is this phenomenon beneficial to a particular species? (2 Marks)

_____

_____

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_____

Q.5. (Total 4 Marks)

Using the given list of organisms, construct a food web. (You may not include all the organisms)

Bats	beetles	spiders	grass	eagles	squirrels	humans	Rabbits
aphids	caterpillars	oak tree	cows	frogs	maize plant	hawks	grass hopper

Q.6. (Total 4 Marks)

a. Who should include single cell proteins in his/her diet? Why? (2 Marks)

_____

_____

_____

_____

b. State TWO conditions under which sedatives are prescribed by a doctor. (2 Marks)

_____

_____

_____

_____

Q.7. (Total 5 Marks)

EITHER

a. As of 2009, 300 million people worldwide were affected by asthma leading to approximately 250,000 deaths per year. Highlighting at least THREE causes of asthma attack, describe the phenomenon of this disease.

OR

b. Dialysis is primarily used to provide an artificial replacement for lost kidney function in people with renal failure. Elaborate the given statement with reference to the composition of the dialysing fluid and its working.

Q.8. (Total 5 Marks)

EITHER

a. With the help of examples, describe the types of movable joints in man.

OR

b. With the help of labelled diagrams only, describe the processes of fragmentation and multiple fission in named organisms.

Q.9. (Total 5 Marks)

EITHER

a. What are the objectives of genetic engineering? With the help of an example justify how genetic engineering is trying to meet its objectives.

OR

b. In order to acquire best protection, it is recommended to receive vaccinations in early childhood. Explain how vaccines can provide best protection against diseases.

END OF PAPER